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Roll No

CS-403 (GS)
B.E. IV Semester Examination, June 2020
Grading System (GS)
Object Oriented Technology
Time : Three Hours

Maximum Marks : 70

Note : i) Attempt any five questions.
ii) All questions carry equal marks.

1. Create a class TIME. Create T1, T2 and T3 as objects of TIME. Write a C++ program to read time T1 and T2 as hh, mm and ss.
Where hh indicates hours, mm indicates minutes and ss indicates seconds. Assume hh, mm and ss all to be integers. Add these times T1 and T2 and store the result in time T3. Print the value of T3.
2. Explain the multiplicity and Navigability with suitable example. Draw the relationship between them?
3. Write a program where numbers are stored in an array in a program. Create two files even and odd using command line argument, where odd file stores all odd numbers in the array and even file stores all even number in the array.
4. a) What is container? What are its different types?
b) Explain following term:
 - i) Persistent objects
 - ii) Files
5. Write a program in Java using class to add two distances entered in feet and inches and store the result in third distance. Pass two distances as argument to a function and the function should return the added distance.

OR

Write short note on: (Any two)

- i) Encapsulation in OOPs
 - ii) Named Association
 - iii) JSP and servlets
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6. What is Encapsulation? What are its advantages? How can encapsulation be enforced in C++?

OR

Discuss important features of C++

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7. Write a program in Java to give the following output.

```
      *
    *  *  *
  *  *  *  *  *
    *  *  *
      *
```

OR

Discuss various types of associations in details.

8. Answer any three of the following:

- a) What is a Meta class?
- b) What do you mean by Static member of a class?
- c) What is Abstract method? What is its utility?
- d) What is a reference?
- e) What do you understand by a friend class?
